University of Rajshahi

Department of Computer Science and Engineering

B.Sc. (Engg.), Part-IV, Odd Semester Examination, 2024
Course: CSE4121 (Object Oriented Design and Design Patterns)
Session:2020-2021, 2019-2020 and 2018-2019
Time: 02 hours Full marks: 35

Answer **Four** questions taking any **Two** questions from each section

Section-A

1. (a) Explain the benefit of representing real-world entities—such as *Citizen*, *MunicipalService*, and *Complaint*—in the Dhaka Urban Management System as objects in your code. 3

   (b) Consider a **Dhaka Urban Management System** scenario: a city resident logs into the smart city portal using their ID and password to verify their identity with the municipal database. The user is then provided options to **report waste collection issues, pay utility bills, request road maintenance**, or **track service status**. Explain how **CRC (Class-Responsibility-Collaborator)** cards can be utilized to isolate and define the system components—including their attributes, functionalities, and interactions—for example, classes such as ***Citizen***, ***MunicipalService***, ***Complaint***, and ***Payment*** can be defined along with their responsibilities and collaborating relationships. 4.75

   (c) What is the difference between software design and software architecture? 1

2. (a) The design principle of **'decomposition'** takes a whole thing and divides it into different parts. There are three types of relationships in decomposition, which define the interaction between the whole and the parts: **association**, **aggregation** and **composition**. Explain those relationships and draw the UML diagram for each of these relationships based on the Dhaka Urban Management System concept. 4.75

   (b) The computation steps taken within a class never need to be known by any other class, as long as they are able to access the interface. What type of encapsulation occurs here? 1

   (c) In the case of an **Agricultural Automation System**, design a parent class **FarmingMachine**, and child classes **IrrigationSystem** and **Harvester**—where the child classes inherit attributes and behaviours from the parent class. Now, explain what type of **design principle** is used here. 3

3. (a) Consider the following code snippet: 3.5

```
public class OrderProcessor {
    ...
    public void process(Order order) {
        order.calculateTotal();
        order.applyDiscount();
        order.generateInvoice();
    }
}
```

   Identify the code smell present in this example. Describe why it is a problem and how you would refactor the code to improve the design.

   (b) What is the Single Responsibility Principle (SRP)? Explain with an example showing violation and correct implementation. 3.5

   (c) How can comments be considered a code smell? Provide examples of when comments indicate poor code quality. 1.75

Section -B

4. (a) What are software design patterns? Explain why learning and applying design patterns is beneficial for software developers. 2.75

   (b) Consider the following `ConfigurationManager` class: 4

```
class ConfigurationManager {
    private String configFilePath;

    public ConfigurationManager(String configFilePath) {
        this.configFilePath = configFilePath;
    }

    public String getProperty(String key) {
        System.out.println("Retrieved property: " + key +
                           " from " + configFilePath);
        return "value";
    }
}
```

   Convert this class into a **Singleton pattern** implementation. Explain why Singleton pattern is appropriate for a configuration manager. Also, demonstrate how to use this singleton class with sample client code.

   (c) Explain the Adapter design pattern with a real-world example. 2

5. (a) Consider the activity of changing TV channels using a remote. 5

   - A user can press corresponding channel number on the remote to change channel.
   - The User can also press up/down button on the remote to change the channel.

   Based on the description above, draw a UML sequence diagram to model the interaction between the user and the TV through the remote.

   (b) What do you understand by conceptual integrity? Explain the ways to achieve conceptual integrity during the design and development of software. 3.75

6. (a) In Facebook, a user can follow another user. Whenever the followed user posts a new status update, all followers receive a notification in their feeds. 4.5

   Which design pattern is most suitable for implementing this notification system? Explain it with a UML class diagram.

   (b) You are building a simple music player that can play only MP3 files. Later, you want the player to support other formats such as MP4 and VLC. The problem is that the library required for MP4 and VLC files works differently from your existing player and does not match the methods your system already uses. You cannot modify the library, and you do not want to rewrite your entire player. 4.25

   Which design pattern can help solve this problem? Explain with an example code snippet.

University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engg. Part-4 Odd Semester Examination-2023
Course: CSE4121 (Object Oriented Design and Design Patterns)

Full Marks: 35 Duration: 2 Hours

[Answer any FOUR questions taking TWO from each section]

SECTION: A

1. a) Explain the benefit of representing real-world entities and in your code as object? 3
   b) Consider an ATM machine scenario: a user inserts their ATM card, enters a PIN to verify their identity with the bank, and is then given options to deposit funds, withdraw cash, or check their account balance. Explain how CRC (Class-Responsibility-Collaborator) cards can be utilized to isolate and define the system components, including their attributes, functionalities, and interactions. 4.75
   c) What is the difference between software design and software architecture? 1

2. a) The design principle of decomposition involves breaking a whole system into smaller, manageable parts. In object-oriented design, the relationships between the whole and its parts are defined by three key types: association, aggregation, and composition. Discuss these three relationships using real-world examples. Draw the corresponding UML diagram for each relationship, clearly labelling the parts and the whole. 6
   b) Consider the following Java implementation of **Whale** class inherited from **Animal**. Can you identify any problem with this implementation? Explain. 2.75

```java
public class Animal {
    private int numberOfLegs;
    private boolean hasTail;

    public Animal(int legs, boolean tail) {
        this.numberOfLegs = legs;
        this.hasTail = tail;
    }

    public void walk() {.. }
    public void run() {. }
    public void eat() {... }
}
```

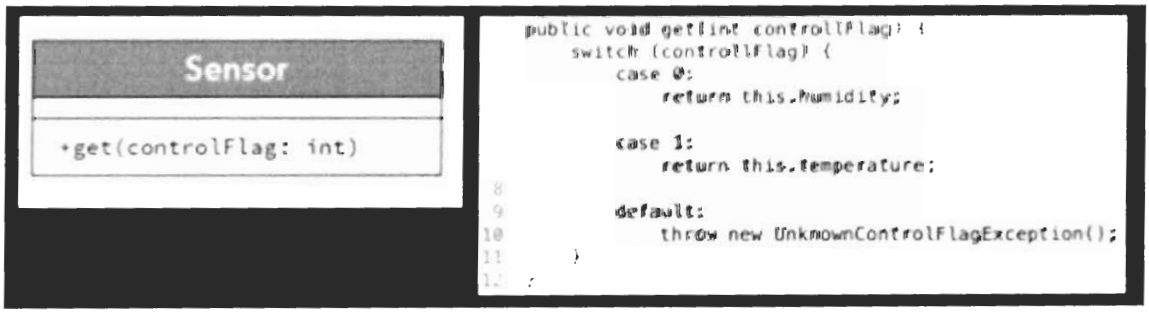
```java
public class Whale extends Animal {
    public Whale () {
        super(0, true);
    }

    private void swim{ ... }

    public void run() {
        this.swim();
    }

    public void walk() {
        this.swim();
    }
}
```

3. Suppose we have a class called **Sensor** that has two purposes, getting humidity and getting temperature sensor readings. Here we have a get method that takes a zero (0) flag if you wanted to return the humidity value, and takes the one (1) flag if you want it to return the temperature value. The class diagram for the Sensor and implementation of its' ***get()*** method is as below:



   a) Evaluate the above design in terms of coupling and cohesion. Give justification in each case. 5
   b) Suggest a better design for above sensor that meets acceptable coupling and cohesion requirement. 3.75

SECTION: B

4. a) What are design patterns, and how do they help solve common design issues in object-oriented software? 2
   b) What are creational design patterns, and why are they important? Give an example of each one. 3
   c) How do structural design patterns contribute to the organization of software? Explain with an example. 3.75

5. a) What is software design pattern and why should we learn them? 1.75
   b) Define creational design patterns and explain their primary purpose in software development. 3
   c) A software system requires a single shared instance of a logging service to record errors across multiple modules. Which creational design pattern should be used? Explain why and outline its structure. 4

6. a) What problem does Adapter design pattern solve? Explain with an example and draw supporting UML diagram. 4.5
   b) Suppose You are designing a system for money withdrawal from bank account using check with the following authorization scenario: 4.25
      - For Tk. < 10,000 only cashier's authorization is sufficient to withdraw money from account.
      - For Tk. 10,000 to 10,00,000 authorization from both cashier and senior officer is needed
      - For Tk > 10,00,000 authorization from Senior officer and Manager is need.

   Implement this behaviour using appropriate software design pattern

University of Rajshahi

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

B.Sc. Engineering Part IV Odd Semester Exam-2022

Course Code: CSE4121 Course Title: Object Oriented Design and Design Patterns

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University of Rajshahi

Time: 02 Hours Full Marks: 35

SECTION A

ANSWER ANY TWO OF THE FOLLOWINGS

1. a) In terms of Object-oriented analysis and design—explain the situations in which a software project may fail. 3
   b) During the process of software design, what are the "Trade-offs between the client's need and Developer's Scope—explain with an example. 4
   c) What Conceptual Mock-ups illustrate in the Software design process? 1.75

2. a) Consider a parent class, and a child class-- where child classes inherit attributes and behaviours of parent classes—explain what type of design principle is appropriate here. 3
   b) The computation steps taken within a class never need to be known by any other class, as long as they are able to access the interface. What type of encapsulation occurs here? Explain with a potential example. 3
   c) Describe the steps of model checking in the software design concept. 2.75

3. a) What is coupling? Describe different couplings with examples. 2
   b) What is the concept of Separation of Concerns in object-oriented design, and how does it contribute to the development of robust and maintainable software systems? 3
   c) Imagine you are tasked with designing a UML sequence diagram for a simple online ordering system. Walk through the steps you would take to create the diagram, including identifying the objects, their interactions, and the order of messages exchanged to illustrate the order processing workflow. 3.75

SECTION B

ANSWER ANY TWO OF THE FOLLOWINGS

4. a) Demonstrate the proxy pattern principle with a UML diagram. 3.5
   b) Decorator design pattern allows additional behaviors or responsibilities to be dynamically attached to an object—explain this situation with a proper example. 3.5
   c) What are the major components of the composite pattern? 1.75

5. a) Consider an executive chef who oversees a large chain of restaurants that serves pasta. Draw a UML diagram to solve this problem in terms of 'Template Method Pattern'. 4.25
   b) A vending machine has several states, and specific actions based on those states, such as purchasing a chocolate bar that costs one dollar. What type of design pattern can be used to design the vending machine software-- explain with a UML diagram. 4.5

6 a) You have been asked by a local coffee shop to create a system that allows managers to see, edit and add employee information. You have decided to create a web application. They have informed you that they may decide to grow and expand later on and would like the system to be flexible to expansion. In this case the MVC pattern should be used. 4

Create a UML class diagram that displays the basic MVC pattern for this web application. The system should keep track of an employee's name, ID number, job title and salary. The controller should be able to get the employee model's properties (getter methods) and change the properties (setter methods). The view should only display employee info.

b) Describe the Dependency Inversion principle? 2.75
c) How can Comments be considered a code smell? 1
d) Look at the code snippet. Which code smell do you detect? 1

```
public class Class1 {
...
public void M(Class2 C) {
   C.doSomething(x);
   C.foo(y);
   C.foo2(z, i);
}
}
```

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
University of Rajshahi
B.Sc. Engineering Part IV Odd Semester Examination-2021
Course Code: CSE4121 Course Title: Object Oriented Design and Design Patterns

Time: 02 Hours Full Marks: 35

SECTION A
[ANSWER ANY TWO OF THE FOLLOWING]

1.(a) What do you understand by Object Oriented Modelling? 1.75

(b) What is the key difference between software design and software architecture? 2.00

(c) Explain how CSR cards are used in modelling a conceptual design with an example. What are the advantages of using CRC cards? 3+2

2.(a) The design principle of **'decomposition'** takes a whole thing and divides it into different parts. There are three types of relationships in decomposition, which define the interaction between the whole and the parts: **association**, **aggregation** and **composition**. Explain those relationships and also draw the UML diagram for each of these relationships. 5.00

(b) Consider an activity of **"changing TV channel using remote".** 3.75

- When a viewer presses a number on the tv remote, the television changes the channel that the tv viewer can see on the tv screen.
- The viewer can also press the up/down button on the remote to change the channel

Use the above description to draw a UML sequence diagram to model the interaction between the user and TV through the remote.

3.(a) What do you understand by Conceptual Integrity? 1.00

(b) Consider the following Smartphone class 2.75

```
public class SmartPhone {
        private byte camera;
        private byte phone;

        public SmartPhone() { … }

        public void takePhoto() { … }
        public void savePhoto() { … }
        public void cameraFlash() { … }

        public void makePhoneCall() { … }
        public void encryptOutgoingSound() { … }
        public void decipherIncomingSound() { … }
}
```

Explain the problems with the class in terms of coupling and cohesion.

(c) How can you apply the **separation of concern** principle to overcome the problems associated with the previous example? Draw a UML diagram of your proposed solution. 5.00

SECTION B
[ANSWER ANY TWO OF THE FOLLOWING]

4.(a) What are "software design patterns", and why should you use them? 2.25

(b) Consider the java class below. Modify the class to convert it to a singleton class. Also, show how you can call the ***executeSql*** method of this singleton class using a demo program. 3.50

```
class Database {
    private String dbName, user, password;

    public Database(String dbName, String user, String password) {
        this.dbName = dbName;
        this.user = user;
        this.password = password;
    }

    public void executeSql(String sql){
        System.out.println("Executed query: "+sql);
    }
}
```

(c) Describe the Adapter pattern and draw its representative class diagram. 3.00

5.(a) Explain how the MVC design pattern can be used to develop a maintainable web application. 4.25

(b) What is the underlying principle of the Observer design pattern? Explain with an example. 4.50

6.(a) Present a scenario where you may need to use a façade design pattern. 1.75

(b) Explain steps to implement façade design pattern 4.00

(c) What are the facilities the façade design pattern provides? 3.00

University of Rajshahi

Department of Computer Science and Engineering

B. Sc. (Engg) Part-II Even Semester Examination 2020

Course: CSE 2211 (Object Oriented Programming and Design Methods)

[Session: 2016-2017]

Full Marks: 52.5 Duration: 3 (Three) Hours

Answer 06 (Six) questions taking any 03 (Three) from each section

Section-A

1. a) Compare Java with C++ as Object Oriented programming language. 2.00

   b) Consider the following java program: 5.00

```
1  public interface IRunable {
2      public void run();
3  }
4
5  Class JavaApp extends ClassA, ClassB implements IRunnable {
6      void main(String *args) {
7          System.in.println("This is a Java program.");
8      }
9  }
```

   Point out the mistakes. Also, correct the program so that it can be compiled.

   c) Mention some differences between array type and **ArrayList** type in java. 1.75

2. a) Write a method called 3.00

```
delete (String str, int m)
```

   that returns the input string with the m-th element removed.

   b) An n-digit number that is the sum of the n-th powers of its digits is called an Armstrong Number. For example, 153 and 370 are Armstrong numbers, since $153 = 1^3 + 5^3 + 3^3$ and $370 = 3^3 + 7^3 + 0^3$. But 15 and 37 are not Armstrong numbers, since $15 \neq 1^2 + 5^2$ and $37 \neq 3^2 + 7^2$. First few Armstrong numbers are 1, 2, 3, 4, 5, 6, 7, 8, 9, 153, 370, 371, 407, 1634 and so on. Write a Java program to find whether the given number is Armstrong or not. 4.50

   c) What are the advantages of java generic classes? 1.25

3. a) What are the uses of **super** keyword in java? 1.75

   b) What do you understand by 'Dynamic method dispatching'? 2.00

   c) What are the conditions for a class to become abstract? Can you declare an object of abstract class? 2.00

   d) Briefly describe the Object class in java. 3.00

4. a) Discuss how **try-catch-finally** block works in java. 3.00

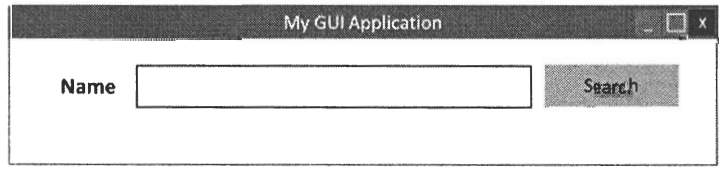
   b) In which cases **FileNotFoundException** is fired by java runtime? 1.25

   c) How can you define your own exception type in java? 2.50

   d) What happens when a class member is declared as static? What restrictions are applied to a method that is declared as static? 2.00

Section-B

5.	a)	What is thread? Discuss Java thread model.	3.25
	b)	Write a program to demonstrate the creation and execution of threads. How can you ensure all threads that started from main must end in order in which they started and main should end in last?	2.50
	c)	What are Java packages? Write the significance of packages.	3.00
6.	a)	What are similarities and differences between java class and java interface?	3.25
	b)	How java garbage collection works? What is finalization?	2.50
	c)	Discuss various ways java implements polymorphism.	3.00
7.	a)	What is the purposes of **File** class in java?	2.25
	b)	What are the differences between **InputStream** class and **Reader** class?	3.00
	c)	Write a simple java program to copy content of a text file into another file.	3.50
8.	a)	Discuss delegate event model with proper diagram.	3.00
	b)	what are the advantages of swing over AWT?	2.00
	c)	Write an AWT based application that uses **FlowLayout** manager to layout the following UI that can be closed by clicking cross [x] icon in the top right corner.	3.75



University of Rajshahi

Department of Computer Science and Engineering

B.Sc. Engg. Part - II, Semester – Even, Examination 2019 (Session: 2016-2017)

Course: CSE-2211 (Object Oriented Programming and Design Methods)

Marks: 52.5 **Duration: 3 hours**

[N.B. Answer **THREE** questions from each section]

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Section –A

1.	a)	What is meant by Object Oriented Programming (OOP)? Write the basic features of OOP.	1.75
	b)	Explain the basic structure of Java Program with example.	5
	c)	Why is main() method static in Java? Explain.	2
2.	a)	How Java garbage collection works? What is finalization?	3.5
	b)	Using example Java code, distinguish between overloading and overriding.	1.75
	c)	Briefly describe how access to member variables and methods are controlled in Java class.	3.5
3.	a)	What do you understand by object-oriented design pattern? Why should we use them?	2.75
	b)	Suppose that your program has a class that you want to have only one instance created and this instance must be reused during the lifetime of your program. Which design pattern will you use to implement that. Explain with example.	3
	c)	Briefly discuss the idea of Observer pattern with example.	3
4.	a)	What are the differences and similarities between abstract class and Interface? Mention a situation when the use of one of them is preferable over the other.	3.5
	b)	See the program below. Identify and explain the problems in the code. After fixing the problem, what will be the output?	3

```
class Base {
    final public void Print() {
        System.out.println("Base");
    }
}

class Derived extends Base {
    public void Print() {
        System.out.println("Derived");
    }
}

class Main{
    public static void DoPrint( Base o ) {
        o.Print();
    }
    public static void main(String[] args) {
        Base x = new Base();
        Base y = new Derived();
        Derived z = new Derived();
        DoPrint(x);
        DoPrint(y);
        DoPrint(z);
    }
}
```

c) What are the purposes of **super** keyword in java? 2.25

Section –B

5 a) Discuss the basic structure of exception handling in Java. 3.5

b) List some common types of exceptions that may occur in Java. 1.75

c) How can you create your own type of exception? Give example 3.5

6. a) briefly explain thread model of Java. 2.75

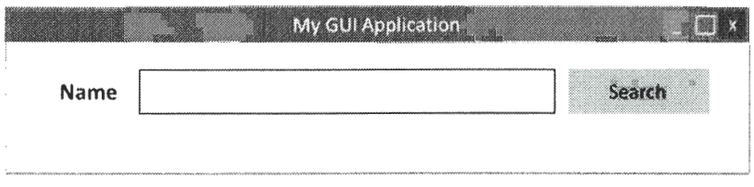
b) How can you create a thread using runnable interface? 3

c) How does interthread communication is performed in Java? 3

7. a) Discuss delegate event model with proper diagram. 3

b) what are the advantages of swing over AWT? 2

c) Write an AWT based application that uses **FlowLayout** manager to layout the following UI that can be closed by clicking cross [x] icon in the top right corner. 3.75



8. a) What is Java package? Why it is useful? 2

b) Explain what problem can arise while using the following generic class. How to fix it? 3

```
1  class Stats<T> {
2    T[] nums;
3
4    Stats(T[] o) {
5      nums = o;
6    }
7
8    double sum() {
9      double x = 0.0;
       for(int i=0; i < nums.length; i++)
11       x += nums[i].doubleValue();
12
13     return x;
14   }
15 }
```

c) Write a java program that reads the content of a file and writes the content back to another file. 3.75